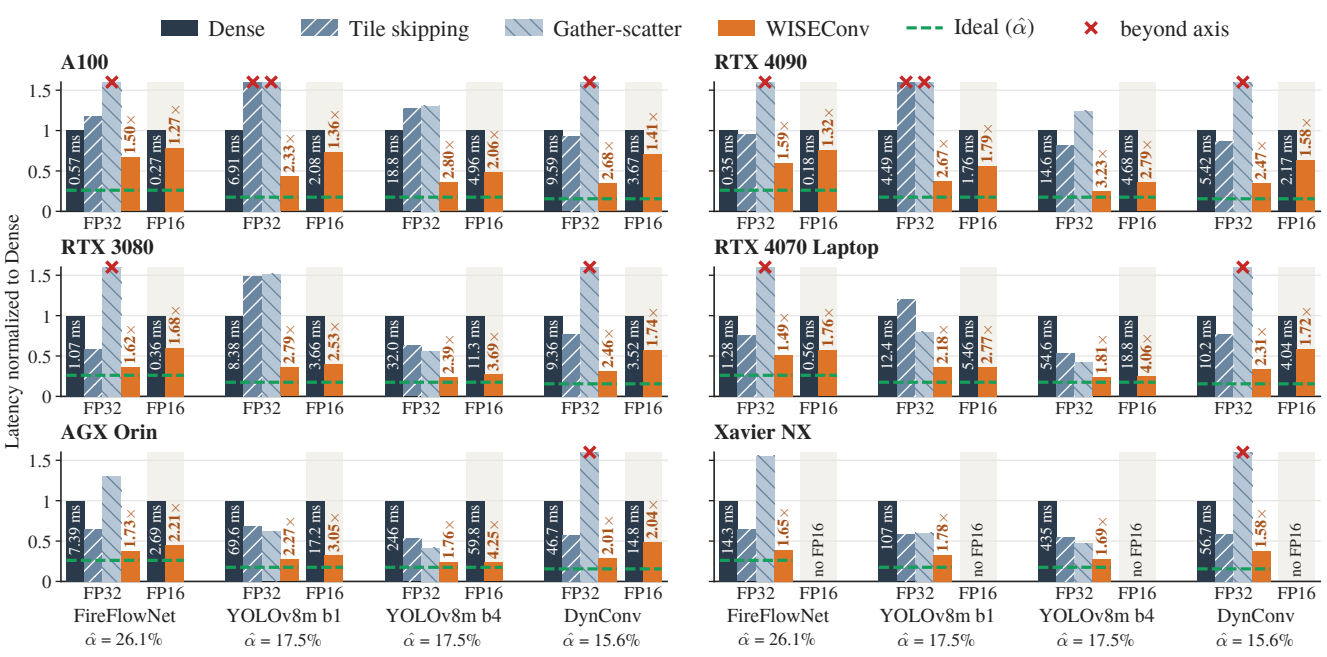
Dense
Tile skipping
Gather-scatter
WISEConv
Ideal ($\hat{\alpha}$)
beyond axis
A100
RTX 4090
RTX 3080
RTX 4070 Laptop
AGX Orin
Xavier NX
Latency normalized to Dense
0.57 ms
1.50×
0.27 ms
1.27×
6.91 ms
2.33×
2.08 ms
1.36×
18.8 ms
2.80×
4.96 ms
2.06×
9.59 ms
2.68×
3.67 ms
1.41×
0.35 ms
1.59×
0.18 ms
1.32×
4.49 ms
2.67×
1.76 ms
1.79×
14.6 ms
3.23×
4.68 ms
2.79×
5.42 ms
2.47×
2.17 ms
1.58×
1.07 ms
1.62×
0.36 ms
1.68×
8.38 ms
2.79×
3.66 ms
2.53×
32.0 ms
2.39×
11.3 ms
3.69×
9.36 ms
2.46×
3.52 ms
1.74×
1.28 ms
1.49×
0.56 ms
1.76×
12.4 ms
2.18×
5.46 ms
2.77×
54.6 ms
1.81×
18.8 ms
4.06×
10.2 ms
2.31×
4.04 ms
1.72×
7.39 ms
1.73×
2.69 ms
2.21×
69.6 ms
2.27×
17.2 ms
3.05×
246 ms
1.76×
59.8 ms
4.25×
46.7 ms
2.01×
14.8 ms
2.04×
14.3 ms
1.65×
no FP16
107 ms
1.78×
no FP16
435 ms
1.69×
no FP16
56.7 ms
1.58×
no FP16
FP32
FP16
FireFlowNet
$\hat{\alpha}$ = 26.1%
YOLOv8m b1
$\hat{\alpha}$ = 17.5%
YOLOv8m b4
$\hat{\alpha}$ = 17.5%
DynConv
$\hat{\alpha}$ = 15.6%